

Project Name: MoneyFlow AI

Smart Financial Management with AI-Driven Insights

App Information

App Name: MoneyFlow AI

Category: Personal Finance / Finance

Version: PowerIn v1.00

Platform: Android

Prmissions Used

MoneyFlow AI is designed with a strong focus on security, privacy, and reliable financial data management. The application uses the following permissions to provide its core functionality.

Essential Permissions

Internet Access (`android.permission.INTERNET`)

- Required for Firebase Authentication, Cloud Firestore, and Firebase Storage services.
- Enables secure synchronization of user accounts, transactions, budgets, and profile information.
- Ensures real-time communication between the application and cloud services.

Network State Access (`android.permission.ACCESS_NETWORK_STATE`)

- Allows the application to detect internet connectivity status.
- Helps manage online and offline data synchronization.
- Provides appropriate user feedback when network connectivity is unavailable.

Security Features

Biometric Authentication (`USE_BIOMETRIC` / `USE_FINGERPRINT`)

- Supports fingerprint and facial recognition authentication.
- Provides an additional security layer alongside the application PIN system.
- Biometric data is processed securely by the Android operating system and is never stored or accessed by MoneyFlow AI.

Privacy Protection

Secure Profile Image Selection

- The application uses Android's secure system image picker for profile image selection.
- Users select specific images directly from their device.
- The application does not require unrestricted access to device storage or gallery content.

Notifications

Smart In-App Alerts

- MoneyFlow AI provides budget-related alerts and notifications using custom in-app user interface components.
- No system-level notification permission is currently required for these alerts.
- Notifications are displayed within the application experience to assist users in monitoring budget limits and financial activity.

Permission Summary

Permission	Purpose	Status
android.permission.INTERNET	Firestore connectivity and cloud synchronization	Required
android.permission.ACCESS_NETWORK_STATE	Network monitoring and offline handling	Required
USE_BIOMETRIC / USE_FINGERPRINT	Biometric security authentication	Optional

Security & Privacy Commitment

MoneyFlow AI does not sell, rent, or share user financial information with third parties. All cloud communications are protected using industry-standard SSL/TLS encryption. User financial records remain associated only with the authenticated account and are protected through Firebase Authentication and secure database access controls.

About the App

MoneyFlow AI is a modern personal finance management application designed to help users gain better control over their financial activities. The application enables users to track income and expenses, manage monthly budgets, and monitor overall financial health through intelligent analytics and visual insights.

- **Problem Solved**

MoneyFlow AI eliminates the complexity of manual expense tracking and provides meaningful financial insights instead of displaying only raw financial data.

- **Target Audience**

- ❖ Students
- ❖ Professionals
- ❖ Freelancers
- ❖ Individuals interested in improving financial management and savings habits

- **Core Purpose**

To help users understand where their money goes and support better financial decision-making.

Main Functionality

- **Real-Time Synchronization**

Secure synchronization of financial data using cloud-based services.

- **Financial Health Monitoring**

Automatically calculates a Financial Health Score based on income and expense patterns.

- **Budget Planning**

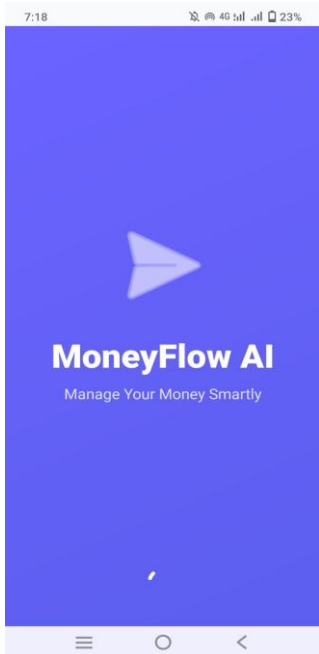
Allows users to define monthly spending limits and monitor spending progress.

- **Smart Insights**

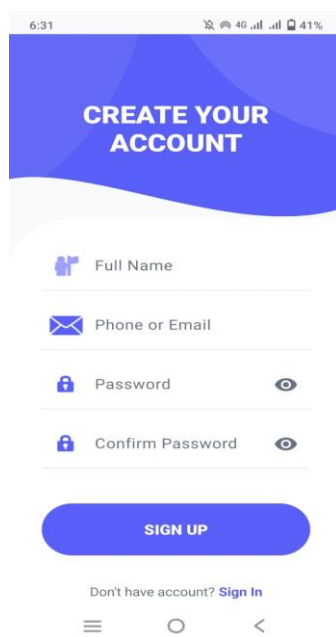
Provides automated insights and recommendations based on spending behavior.

App Screenshots

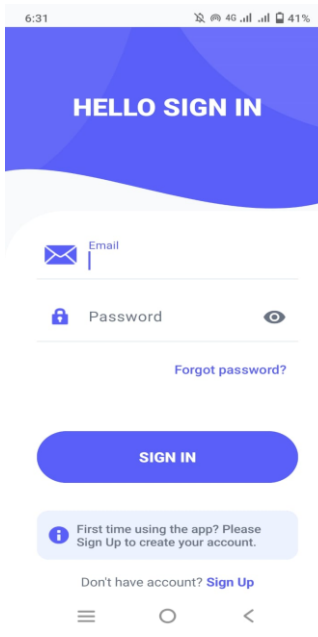
Splash Screen



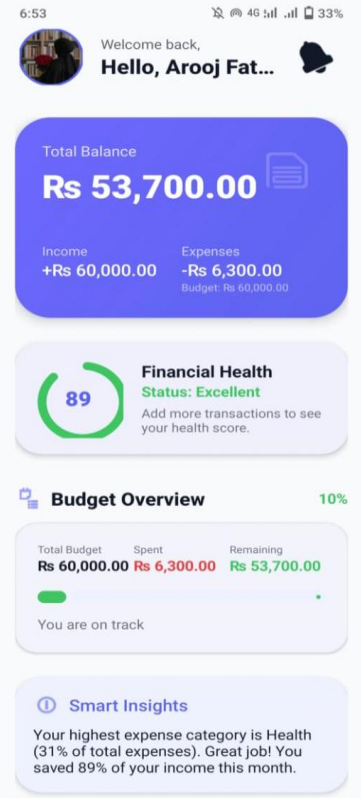
Sign Up Screen



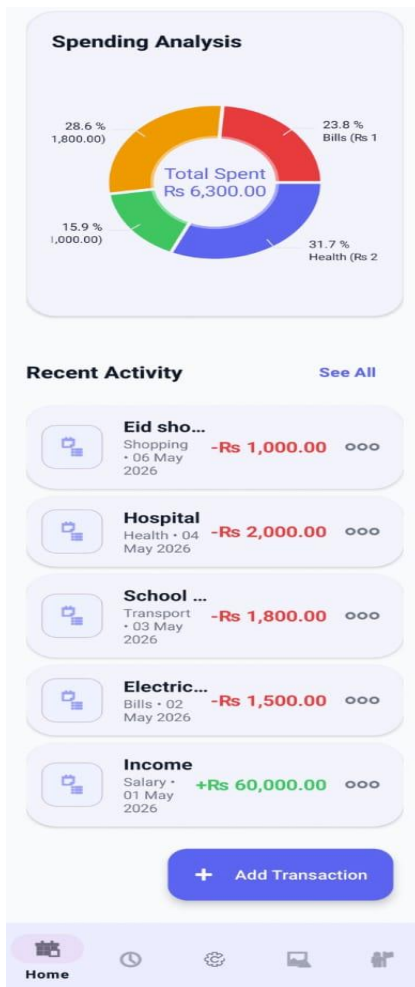
Login Screen



Dashboard Screen 1



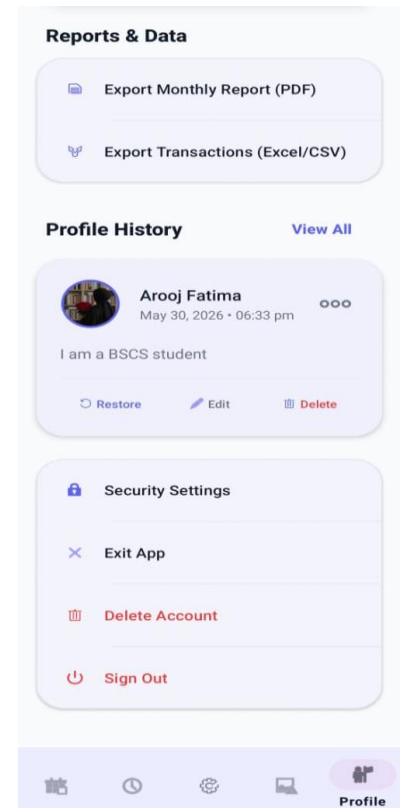
Dashboard Screen 2



Profile Screen 1



Profile Screen 2



App Screens

- **Splash Screen & Onboarding**
Provides an introduction to the application's key features and user experience.
- **Login & Signup**
Allows secure user registration, authentication, and password recovery.
- **Dashboard**
Displays financial summaries, charts, health scores, and recent transactions.
- **Transactions Screen**
Provides a complete and searchable record of all financial activities.
- **Budget Manager**
Enables users to set budgets and track category-wise spending.
- **Add Transaction Screen**
Allows users to quickly record income and expense entries.
- **Profile & Settings**
Manages user profile information, security settings, and preferences.
- **PIN Security Screen**
Provides additional protection using a secure application-level PIN.

Features

- User Registration and Login
- Firebase Authentication
- Income and Expense Tracking
- Financial Dashboard
- Budget Planning and Monitoring
- Smart Financial Insights
- Interactive Expense Charts
- Transaction Management (Create, Read, Update, Delete)
- Profile Management
- Profile History Tracking
- Currency Customization
- PIN Security Protection
- Cloud Data Synchronization

Technologies Used

- **Programming Language**
Java

- **Architecture**
MVVM (Model-View-ViewModel)
- **Backend Services**
Firebase Authentication
Cloud Firestore
Firebase Storage
- **Local Database**
Room Persistence Library
- **User Interface & Libraries**
Material Design 3
MPAndroidChart
Lottie Animations
Glide Image Loading Library
- **Navigation**
Android Jetpack Navigation Component

APK Download

The APK file is included in this project submission and can be found in the following folder:

MoneyFlow_project/apk/MoneyFlow.apk

Download the APK file from the apk folder and install it on an Android device to use the application.

APK Installation Guide

1. Download the MoneyFlowAI.apk file.
2. Enable “Install from Unknown Sources” in Android Settings if required.
3. Tap the APK file and select **Install**.
4. Wait for the installation process to complete.
5. Launch the application and create an account.

Project Setup and Execution

1. Download or extract the project source code.
2. Open Android Studio.
3. Select **Open Existing Project**.
4. Open the MoneyFlow AI project folder.
5. Allow Gradle synchronization to complete.
6. Ensure Firebase configuration is properly connected.

7. Connect an Android device or launch an emulator.
8. Run the application.

Demo Video

A complete demonstration video of the MoneyFlow AI application has been recorded and uploaded separately for reference.

The video explains:

- Application workflow
- Login and Signup process
- Dashboard functionality
- Transaction management
- Budget and profile features

Video Location:

The demo video is provided separately in the project submission folder outside this PDF. It is not included inside the document and can be accessed directly from the submitted project files.

Users can view the video to understand the complete working flow of the application.

Privacy and Security Features

The Privacy Policy document is included in this project submission and can be found in the following location:

MoneyFlow project/docs/privacy_policy.pdf

Please refer to this document for detailed information regarding data collection, storage, usage, and security practices implemented in the application.

Future Enhancements

- AI-Based Expense Categorization
- Automated Bill Reminder System
- Export Reports to PDF and CSV
- Shared Family Wallets
- Advanced Financial Analytics
- Enhanced Budget Recommendations

- Multi-Device Synchronization Improvements

Developer Information

Developed By:

Arooj Fatima

Role: Lead Developer & Implementation

Roll Number: UL-BSCSM-B-23-16

Responsible for the complete development, coding, integration, testing, and implementation of the MoneyFlow AI application.

Afeera Hameed

Role: Contribution to Project Idea & Feature Suggestions

Roll Number: UL-BSCSM-B-23-68

Contributed to the project idea and provided suggestions for app features.

Academic Information

Degree Program: Bachelor of Science in Computer Science (BSCS)

Semester: 6th Semester

Session: Section B (M)

Project Type: Semester Project

Conclusion

MoneyFlow AI is a comprehensive financial management solution that combines budgeting, transaction tracking, cloud synchronization, and intelligent financial insights in a secure and user-friendly Android application.